

C++ FOR QUANTS

If your C++ is written like Python,
You are not writing low-latency code.

Save this before you learn the wrong C++

MOST PEOPLE SAY: “I KNOW C++.”

Low-latency roles expect:

- predictable performance
- memory discipline
- cache awareness

This is not standard C++.

FIRST: FOUNDATIONAL C++

Before low latency, you must be solid in:

- object lifetimes
- RAI
- move semantics
- const-correctness
- references vs pointers

If these are weak, stop here.

CORE C++ BOOKS

Read slowly. Re-read.

- Effective Modern C++ – Scott Meyers
- The C++ Programming Language – Bjarne Stroustrup
- C++ Concurrency in Action – Anthony Williams

This is the baseline.

PERFORMANCE & MEMORY

Low latency lives here.

- stack vs heap allocation
- cache lines & false sharing
- alignment & padding
- branch prediction

Resources:

- What Every Programmer Should Know About Memory – Ulrich Drepper
- Optimized C++ – Kurt Guntheroth



SYSTEMS THINKING

C++ alone is not enough.
You must understand:

- OS scheduling
- NUMA effects
- CPU pinning
- syscall cost

Ignoring the OS = unstable latency.

CONCURRENCY & LOCK-FREE BASICS

Low-latency systems minimise blocking.

Learn:

- atomics
- memory ordering
- lock-free queues
- contention trade-offs

Resource:

- C++ Concurrency in Action (focus on atomics)

PAPERS & ARCHITECTURES TO STUDY

Read for ideas, not code.

- LMAX Disruptor architecture papers
- Memory Barriers – Ulrich Drepper
- Intel Latency Optimisation Guides
- Linux perf & scheduler docs

This is how real systems are built.

OPEN-SOURCE CODE



Study implementation details.

- Boost.Lockfree
- Folly (Meta)
- Aeron messaging concepts
- C++ Disruptor implementations

Focus on design choices, not syntax.

WHAT TO AVOID

→ Avoid:

- generic C++ tutorials
- STL abuse without cost awareness
- “low-latency” YouTube clickbait
- benchmarking averages instead of p99+

These waste time.

FINAL TAKEAWAY



Low-latency C++ is:

- systems thinking
- memory discipline
- performance skepticism

Not syntax.

Not design patterns.

SAVE THIS IF YOU'RE TARGETING HFT OR INFRA ROLES.

→ Comment C++ if you want a
Practice roadmap with projects IDEAS.